

Predicting the Second Gift

Which first-time donors will give again, and which ones a small nonprofit should spend staff time on.

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STAKEHOLDER

Development lead at a ~\$500K nonprofit, no data staff

DECISION

Which first-time donors get a personal follow-up, and where the cutoff sits

DATA

DonorsChoose Open Data 2002–2019 (ICPSR 37898): 11.4M donations, 2.1M projects

GROUND TRUTH

Observed second gift within 12 months, held-out 2017–18 cohorts

HONEST BASELINE

RFM — the recency-and-amount rule small nonprofits actually use

01 PROBLEM AND BUSINESS APPLICATION

Most nonprofits acquire a donor once and never hear from them again. The second gift is where retention is won or lost, and small organizations have the least capacity to work it. Our stakeholder is the development lead at a nonprofit with roughly a \$500K budget and no data staff. Each month she can personally follow up with only a fraction of first-time donors, and today she builds that list by hand from recency and gift size.

The decision: given a fixed outreach budget in staff hours, which first-time donors get a follow-up, and where does the cutoff sit? Our output is a ranked follow-up list with a recommended threshold and the expected dollars behind it. One of us founded GoGood Technologies, a donor-engagement platform whose customers are this stakeholder, so practitioners can check the output. Deployed, it replaces the hand-built list and puts a number on the budget question.

02 DATASET AND SOURCE

DonorsChoose Open Data, United States, 2002–2019 (ICPSR 37898). Public-use files, free with a no-cost ICPSR account. Two files: Donations (11,377,479 records) and Projects (2,149,817), covering September 2002 to June 2019 with activity through December 2019. Donations carries DONOR_ID, so we reconstruct each donor's history and build our own label, plus amount, month, donor type, and matched, teacher-referred, and thank-you-packet flags. Two constraints accepted up front: dates are month-level, and there is no demographics file, so donor type is the only donor attribute.

03 WHAT WE WILL BUILD

- **Label.** For each donor whose first recorded gift falls in the observation window: a second gift within 12 months? Only cohorts whose window closes before December 2019 are labeled.
- **Features.** Donation-side (amount, seasonality, matched, gift card, teacher-referred, donor type, thank-you packet) and project-side (subject, grade, cost, state). Fields that could post-date the second gift are checked for timing before use.
- **Split.** Time-based, never random: train on first-gift cohorts through 2016, hold out 2017 and 2018, observe outcomes through December 2019.
- **Decision layer.** Predicted probability and second-gift amount become expected value per contact, minus an explicit cost per contact in staff time; the threshold is derived from those payoffs, not tuned.
- **Cost at scale.** Scoring a year of first-time donors is a batch job on one machine, with no API spend. The operating cost is the staff hours the threshold allocates, and we report it.

04 EVALUATION — WHAT WE MEASURE, AGAINST WHAT GROUND TRUTH

Ground truth is observed donor behavior in the held-out cohorts, not hand labels. Four measurements:

- **Discrimination and calibration.** PR-AUC because the class is imbalanced, plus a calibration curve: an uncalibrated score cannot support a threshold.
- **Honest baselines.** RFM, the recency-and-amount heuristic small nonprofits actually use, plus random targeting and contact-everyone. Business metric: net value retained per hour of outreach at a fixed budget. If the model does not beat RFM, that is the finding.
- **Does stewardship predict retention?** The thank-you-packet flag records a real post-gift action. We estimate its association with a second gift controlling for gift size; packets were not randomly assigned, so association, not cause.
- **Error analysis by cohort.** We expect the model to be weakest on donors with the thinnest history, the case the stakeholder cares about most.

Limitation we test rather than assume. DonorsChoose donors are marketplace donors; small-nonprofit donors are relational. We test which signal survives: behavioral features versus platform-specific ones.

05 WHAT WE NEED FROM YOU

- Confirmation that a well-supported negative finding against the RFM baseline is an acceptable result.
- The two files total about 13 GB, more than JupyterHub comfortably holds. Either guidance on working with them there, or approval to use a stratified sample of donor cohorts.
- Any concern about anchoring the framing to a company a team member founded.